أكاديمية طويق
Tuwaiq Academy

Hotel Management System

Unit 1 project

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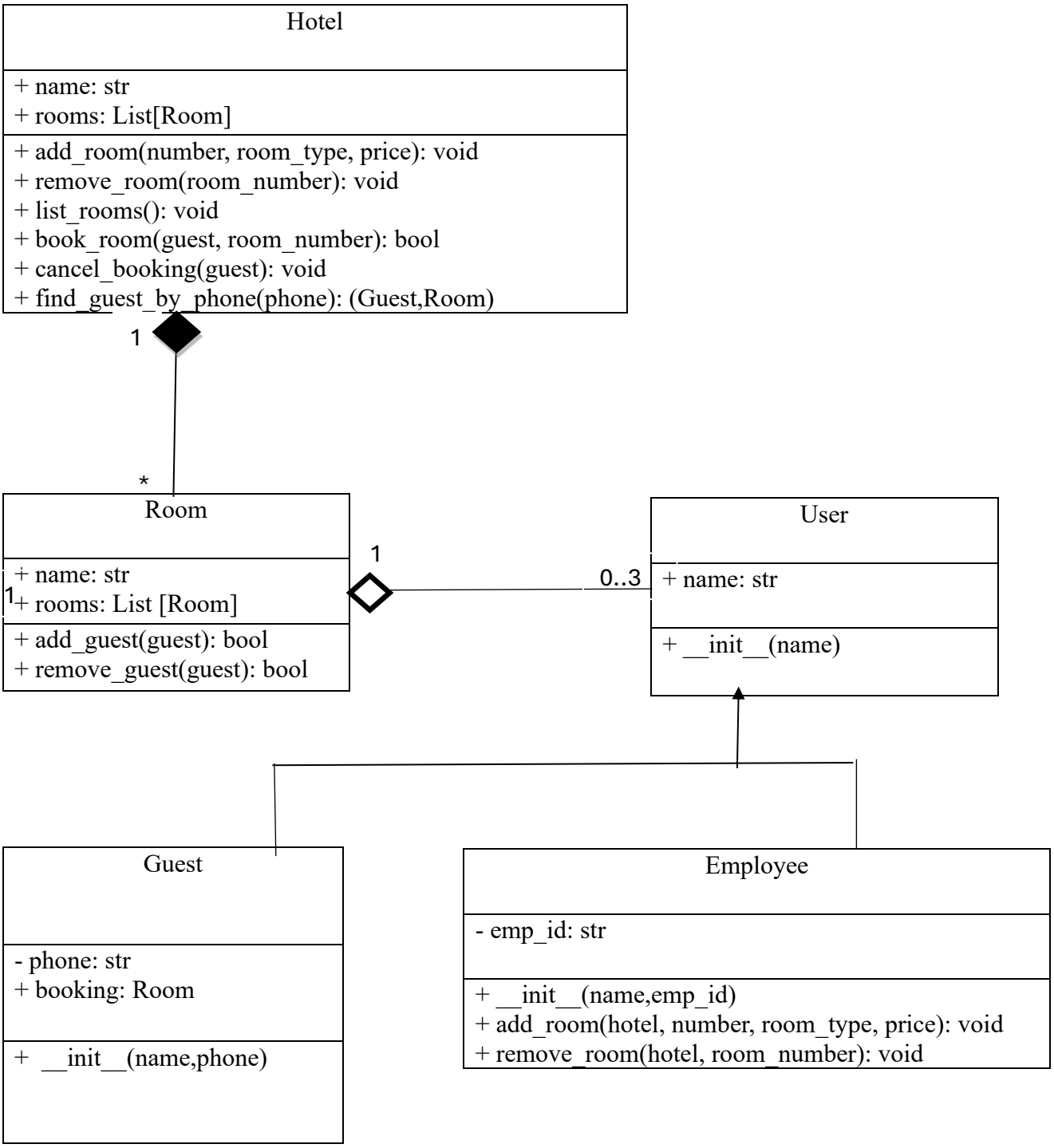
Introduction:

The project aims to build a hotel reservation management system consisting of two main users. The first user is the employee, who is responsible for managing the hotel and adding rooms. The second user is the guest, who can book a room at the hotel and has the freedom to cancel the reservation if needed. When the guest performs a check-out, the reservation details are saved in a file called history.json, which stores the guest's name, phone number, number of nights stayed, and the total cost of the stay.

Solution Design (UML):



Classes Description

Hotel Class

- Attributes:
  - name: the name of the hotel.
  - rooms: list of Room objects in the hotel.

- Methods:
  - __init__(name: str): constructor that initializes the hotel with a name and empty rooms list.
  - add_room(number: int, room_type: str, price: float): adds a new room to the hotel.
  - remove_room(room_number: int): removes a room from the hotel by its number.
  - list_rooms(): prints all rooms with their details.
  - book_room(guest: Guest, room_number: int) -> bool: books a room for a guest; returns True if successful.
  - cancel_booking(guest: Guest): cancels a guest's booking.
  - find_guest_by_phone(phone: str) -> (Guest, Room): finds a guest and their room by phone number.

Room Class

- **Attributes:**
  - number: the room number.
  - room_type: type of the room (e.g., VIP, Normal).
  - price: price of the room per night.
  - status: availability status ("Available" or "Full").
  - guests: list of Guest objects currently in the room.

- **Methods:**
  - __init__(number: int, room_type: str, price: float): constructor that initializes a room.
  - add_guest(guest: Guest) -> bool: adds a guest to the room; returns True if successful.

- remove_guest(guest: Guest) -> bool: removes a guest from the room; returns True if successful.

User Class

- **Attributes:**
  - name: the name of the user.

- **Methods:**
  - __init__(name: str): constructor that initializes the user with a name.

Guest Class

- **Attributes:**
  - phone: the guest's phone number (private).
  - booking: the Room object the guest has booked (default None).

- **Methods:**
  - __init__(name: str, phone: str): constructor that initializes a guest with name and phone.

Employee Class

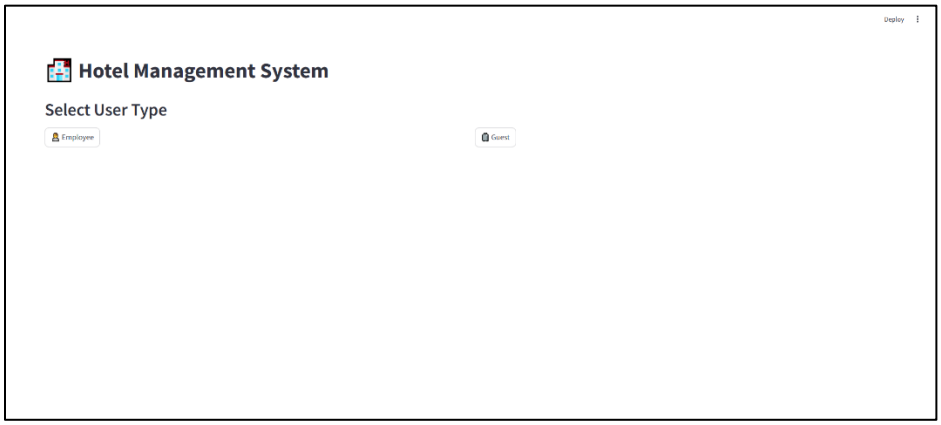
- **Attributes:**
  - emp_id: the employee's ID (private).

- **Methods:**
  - __init__(name: str, emp_id: str): constructor that initializes an employee.
  - add_room(hotel: Hotel, number: int, room_type: str, price: float): adds a room to the hotel.
  - remove_room(hotel: Hotel, room_number: int): removes a room from the hotel.

Streamlit

Streamlit is used in this project to provide a simple and interactive user interface for the hotel management system. It allows employees and guests to log in, navigate menus, view, add, search, sort, book, cancel, and check out rooms, all through buttons, inputs, and sidebars.

1



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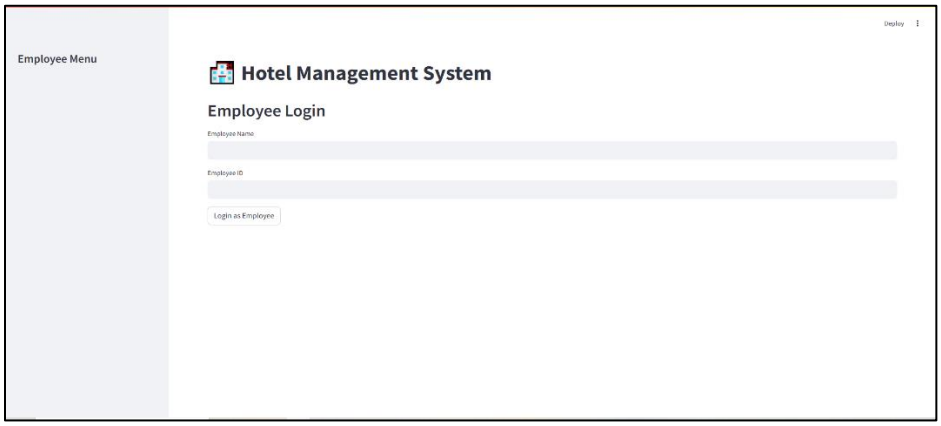

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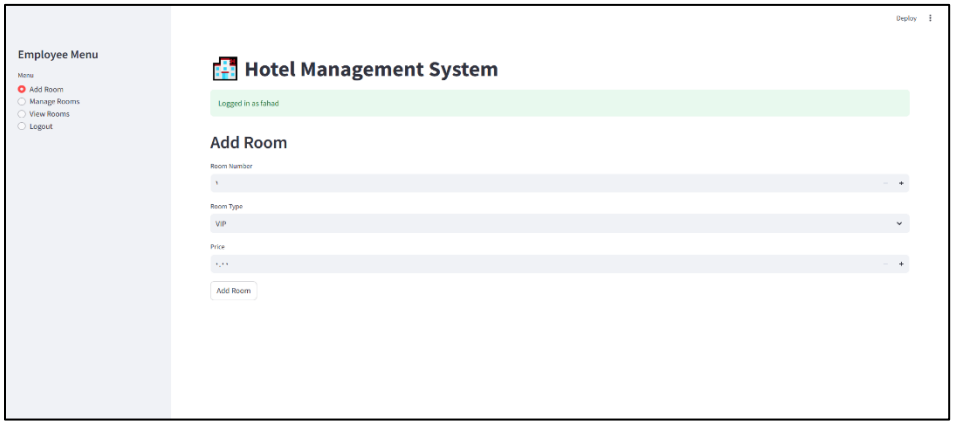

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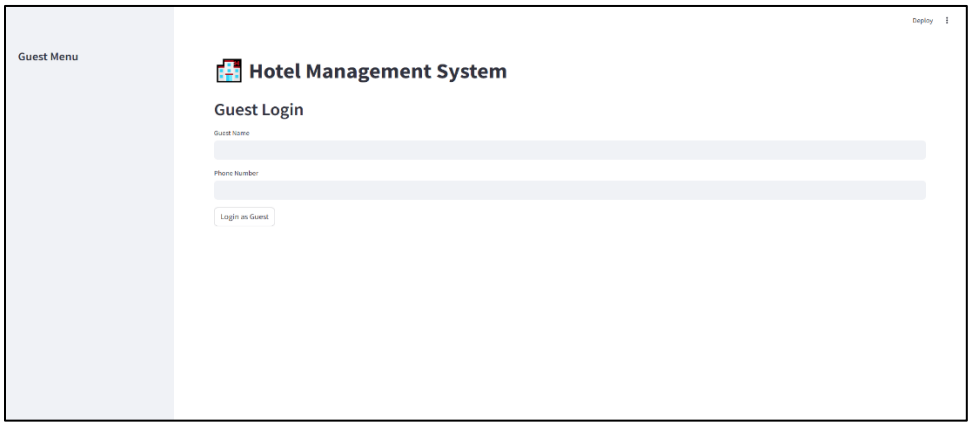
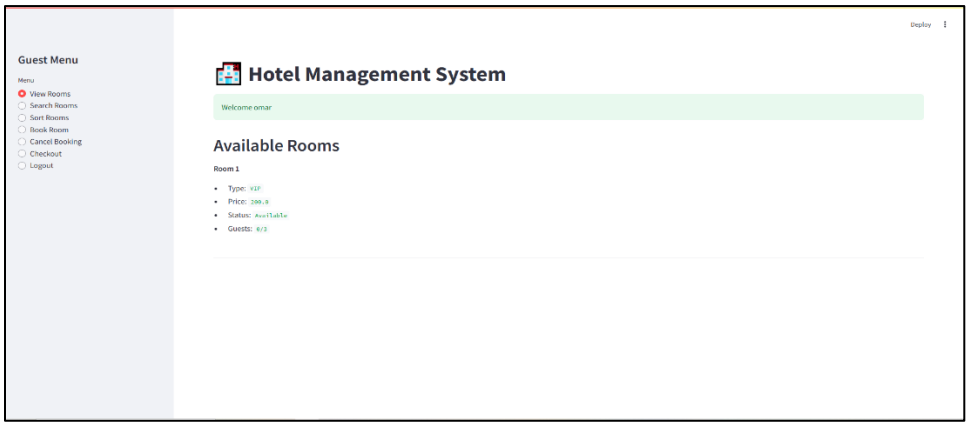

Deploy
Guest Menu
Hotel Management System
Guest Login
Guest Name
Phone Number
Login as Guest

5


Deploy
Guest Menu
Menu
View Rooms
Search Rooms
Sort Rooms
Book Room
Cancel Booking
Checkout
Logout
Hotel Management System
Welcome omar
Available Rooms
Room 1
Type: VIP
Price: 200.0
Status: Available
Guests: 0/3